

Network Security

NSLookup, Dig, and DNS Poisoning

Introduction

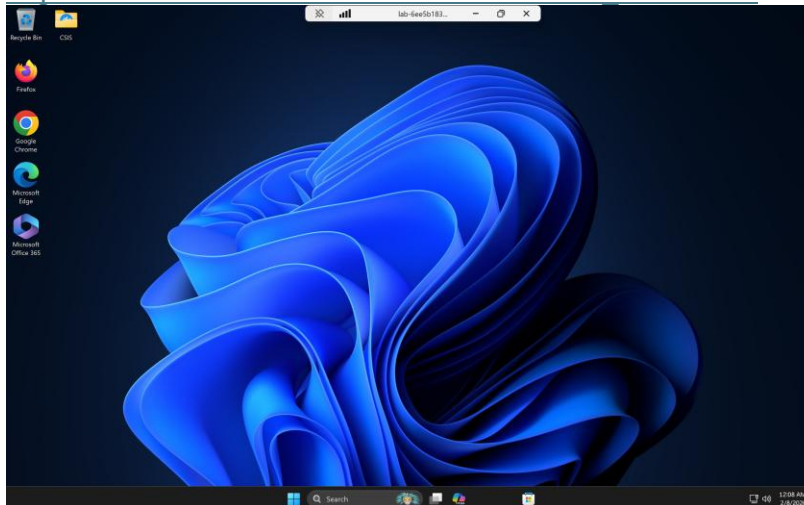
This lab is designed to understand Domain Naming System (DNS) resolution mechanism and practically how DNS can be exploited as an attack vector in cybersecurity environment. *nslookup* tool is used to analyze DNS resolutions in Ubuntu environment. In this lab, *nslookup* utility was installed and used to resolve domain names such as google.com, mnstate.edu, wku.edu, and amazon.com into their corresponding IP addresses. The *dig* utility was installed and used to get more details such as DNS headers, flags, query types, TTL values, and response records. This information gives deep understanding about DNS queries. The final stage of the lab, the local host file was modified to start a DNS poisoning attack. By manually mapping a legitimate domain name to an incorrect IP address can redirect users to unintended destinations.

This lab demonstrates both normal DNS operations and malicious DNS manipulation. DNS is critical component and frequently targeted by the attackers for redirection, and man-in-the-middle attacks. This lab provides the importance of DNS awareness with practical experience with real-world attack scenario.

Step 1 - Verifying/Installing Ubuntu over Windows Subsystem for Linux (WSL)

1. Logged to Azure Lab environment using following web link and logged into the Windows VM using Oracle VirtualBox.

https://labs.azure.com/virtualmachines?feature_vnext=true



2. On Windows, opened PowerShell as an administrator and checked the installed *wsl* distribution.

```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\CSIS-Admin> wsl --list
Windows Subsystem for Linux Distributions:
docker-desktop (Default)
Ubuntu
PS C:\Users\CSIS-Admin> |
```

Step 2 – NSLOOKUP

1. The *nslookup* utility was not preinstalled on Ubuntu. Then installed required utility before starting to work with *nslookup*.

```
hashan@lab8K4EB1: ~$ nslookup google.com
Command 'nslookup' not found, but can be installed with:
sudo apt install bind9-dnsutils
hashan@lab8K4EB1:~$ sudo apt install bind9-dnsutils
[sudo] password for hashan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

2. Ran *nslookup* on URL *google.com* and captured the result.

```
hashan@lab8K4EB1:~$ nslookup google.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 209.85.145.138
Name:   google.com
Address: 209.85.145.139
Name:   google.com
Address: 209.85.145.100
Name:   google.com
Address: 209.85.145.102
Name:   google.com
Address: 209.85.145.101
Name:   google.com
Address: 209.85.145.113
Name:   google.com
Address: 2607:f8b0:4001:c74::71
Name:   google.com
Address: 2607:f8b0:4001:c74::65
Name:   google.com
Address: 2607:f8b0:4001:c74::66
Name:   google.com
Address: 2607:f8b0:4001:c74::8a

hashan@lab8K4EB1:~$
```

For this search the DNS query was resolved using DNS server 10.255.255.254 on port 53. There are multiple IPv4 and IPv6 addresses found from nslookup search. Google is a popular website, and a lot of people use it, so google uses load balancing and distributed DNS infrastructure to improve availability and performance. The non-authoritative means that results were provided by a caching DNS server rather than Google's authoritative name server.

3. Ran *nslookup* on URL mnstate.edu and captured the result.

```
hashan@lab8K4EB1:~$ nslookup mnstate.edu
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   mnstate.edu
Address: 199.17.127.37

hashan@lab8K4EB1:~$ |
```

The DNS query was resolved using DNS server 10.255.255.254 on port 53. The nslookup command resolved the domain name to one IPv4 address 199.17.127.37. Minnesota State University Moorhead is using one IPv4 address to provide their web services to clients. Because it is not required to have many IPv4 addresses to support their clients.

4. Ran *nslookup* on URL wku.edu and captured the result.

```
hashan@lab8K4EB1:~$ nslookup wku.edu
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   wku.edu
Address: 161.6.96.74

hashan@lab8K4EB1:~$
```

This search also resolved DNS query using DNS server 10.255.255.254 on port 53. Western Kentucky University also like MSUM, because it is also using one IPv4 address 161.6.96.74 to provide its web services to clients. This is sufficient for the institution's traffic requirements and eliminates the need for multiple IPv4 addresses or extensive load balancing.

5. Ran *nslookup* on URL amazon.com and captured the result.

```
hashan@lab8K4EB1:~$ nslookup amazon.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   amazon.com
Address: 98.87.170.71
Name:   amazon.com
Address: 98.82.161.185
Name:   amazon.com
Address: 98.87.170.74

hashan@lab8K4EB1:~$ |
```

This search also resolved DNS query using DNS server 10.255.255.254 on port 53. Amazon is also same as Google which provides facilities for many clients. Amazon also uses load balancing and distributed DNS infrastructure to improve availability and performance by having multiple IPv4 addresses.

Step 3 – DIG

1. Installed *dig* utility using *sudo apt install dnsutils* command.
2. Ran *dig* on URL google.com and captured the result.

```
hashan@lab8K4EB1:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31171
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1224
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 14      IN      A       173.194.206.102
google.com.                 14      IN      A       173.194.206.101
google.com.                 14      IN      A       173.194.206.138
google.com.                 14      IN      A       173.194.206.100
google.com.                 14      IN      A       173.194.206.113
google.com.                 14      IN      A       173.194.206.139

;; Query time: 0 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Sun Feb 08 14:08:52 CST 2026
;; MSG SIZE rcvd: 135

hashan@lab8K4EB1:~$
```

The result shows successful DNS response with status NOERROR. The Header shows a DNS query operation with a unique transaction ID used to match response to the user. The flag *qr* indicates this message is a response, not a request, *rd* flag indicates that the client requested that the DNS server perform recursive resolution, and *ra* flag confirms that the DNS server supports recursive queries. The UDP payload size of 1224 bytes, allowing larger DNS responses without requiring TCP.

The answer section contains multiple IPv4 (A) records with a very low Time to Live (TTL) value of 14 seconds, which shows that Google uses of load balancing and dynamic traffic management. The DNS response was received using UDP port 53 from the DNS server 10.255.255.254, with a total message size of 135 bytes.

3. Ran *dig* on URL mnstate.edu and captured the result.

```
hashan@lab8K4EB1:~$ dig mnstate.edu

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> mnstate.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62938
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1224
;; QUESTION SECTION:
;mnstate.edu.              IN      A

;; ANSWER SECTION:
mnstate.edu.               428     IN      A       199.17.127.37

;; Query time: 10 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Sun Feb 08 14:10:03 CST 2026
;; MSG SIZE rcvd: 56

hashan@lab8K4EB1:~$
```

This query is also success with NOERROR. The answer section contains one IPv4 (A) records with a TTL of 428 seconds, resolving to IPv4 address 199.17.127.37. This high TTL value indicates a stable DNS configuration with less frequent IP rotation. The DNS response was received using UDP port 53 from the DNS server 10.255.255.254, with a total message size of 56 bytes.

4. Ran *dig* on URL wku.edu and captured the result.

```
hashan@lab8k4EB1:~$ dig wku.edu

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> wku.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44301
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1224
;; QUESTION SECTION:
;wku.edu.                IN      A

;; ANSWER SECTION:
wku.edu.                 60      IN      A      161.6.96.74

;; Query time: 50 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Sun Feb 08 14:11:00 CST 2026
;; MSG SIZE rcvd: 52

hashan@lab8k4EB1:~$
```

This query is also success with NOERROR. The answer section contains one IPv4 (A) records with a TTL of 60 seconds, resolving to IPv4 address 161.6.96.74. The DNS response was received using UDP port 53 from the DNS server 10.255.255.254, with a total message size of 52 bytes.

5. Ran *dig* on URL amazon.com and captured the result.

```
hashan@lab8k4EB1:~$ dig amazon.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> amazon.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49033
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1224
;; QUESTION SECTION:
;amazon.com.             IN      A

;; ANSWER SECTION:
amazon.com.             900     IN      A      98.82.161.185
amazon.com.             900     IN      A      98.87.170.71
amazon.com.             900     IN      A      98.87.170.74

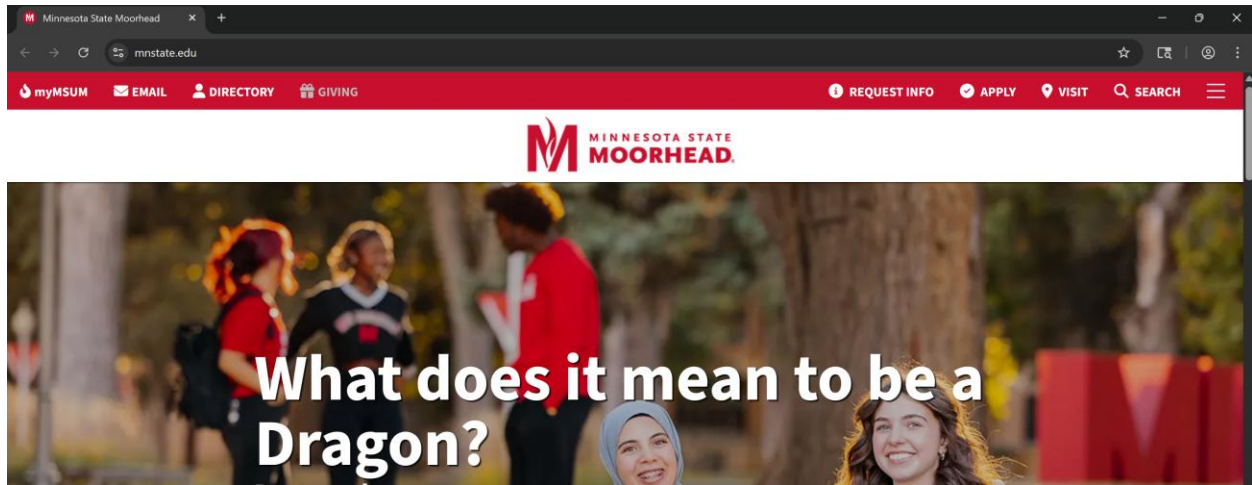
;; Query time: 20 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Sun Feb 08 14:11:28 CST 2026
;; MSG SIZE rcvd: 87

hashan@lab8k4EB1:~$
```

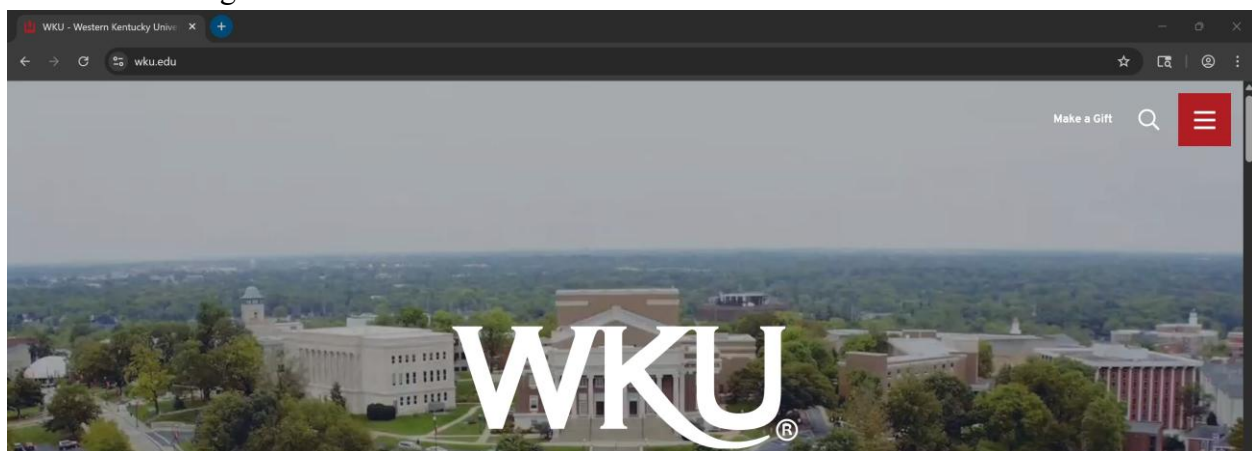
This query is also success with NOERROR. The answer section contains one IPv4 (A) records with a TTL of 900 seconds. The Amazon uses of multiple A records for redundancy and scalability to provide services to their clients, while maintaining a longer TTL compared to Google. The DNS response was received using UDP port 53 from the DNS server 10.255.255.254, with a total message size of 87 bytes.

Step 4 – DNS Poisoning

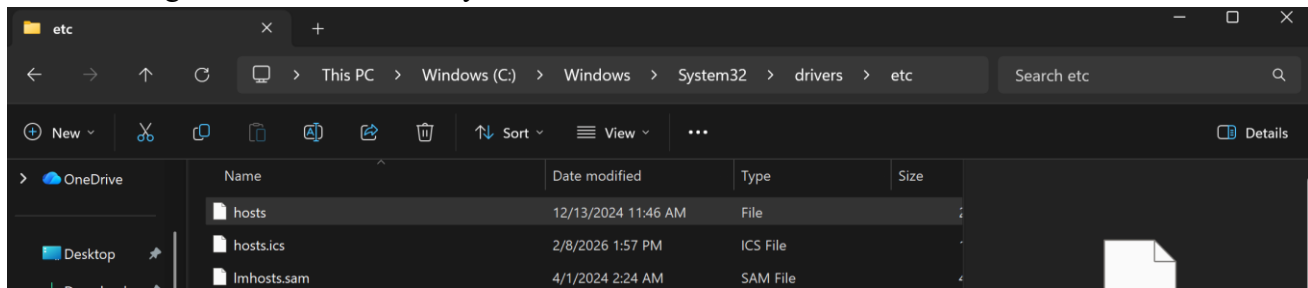
1. Opened google browser and went to Minnesota State University Moorhead website using URL www.mnstate.edu.



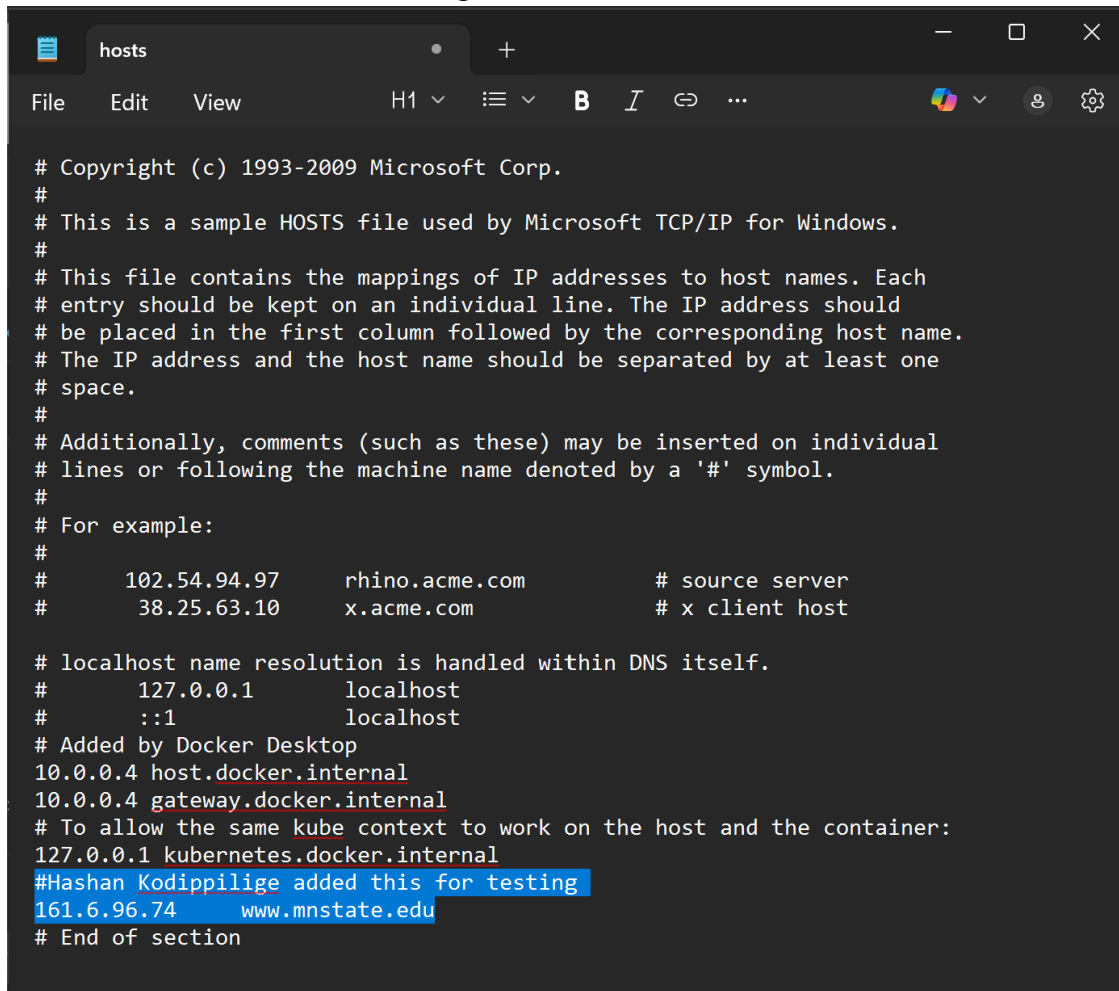
2. In a web browser, used IPv4 address 161.6.96.74 and it entered directly into the Western Kentucky University website www.wku.edu. This confirms that the IP address belongs to Western Kentucky University's web server and it correctly mapped to the WKU domain through DNS and web server configuration.



3. Navigated to C:\Windows\System32\drivers\etc and found the host file.



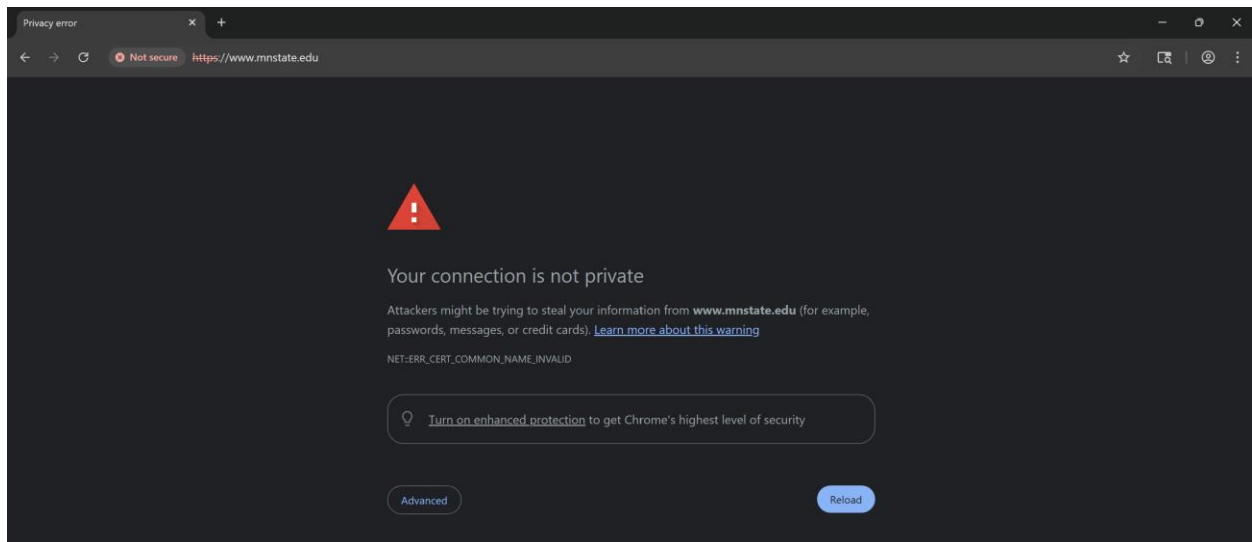
4. Then the host file was opened using Notepad with administrator privileges and added two lines as follows and saved the changes.



```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com           # source server
#       38.25.63.10       x.acme.com               # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
# Added by Docker Desktop
10.0.0.4 host.docker.internal
10.0.0.4 gateway.docker.internal
# To allow the same kube context to work on the host and the container:
127.0.0.1 kubernetes.docker.internal
#Hashan Kodippilige added this for testing
161.6.96.74 www.mnstate.edu
# End of section
```


5. After saving the changes, access to www.minstate.edu and browser security warnings appear due to a certificate mismatch. It demonstrates that DNS poisoning can redirect users to unintended destinations.

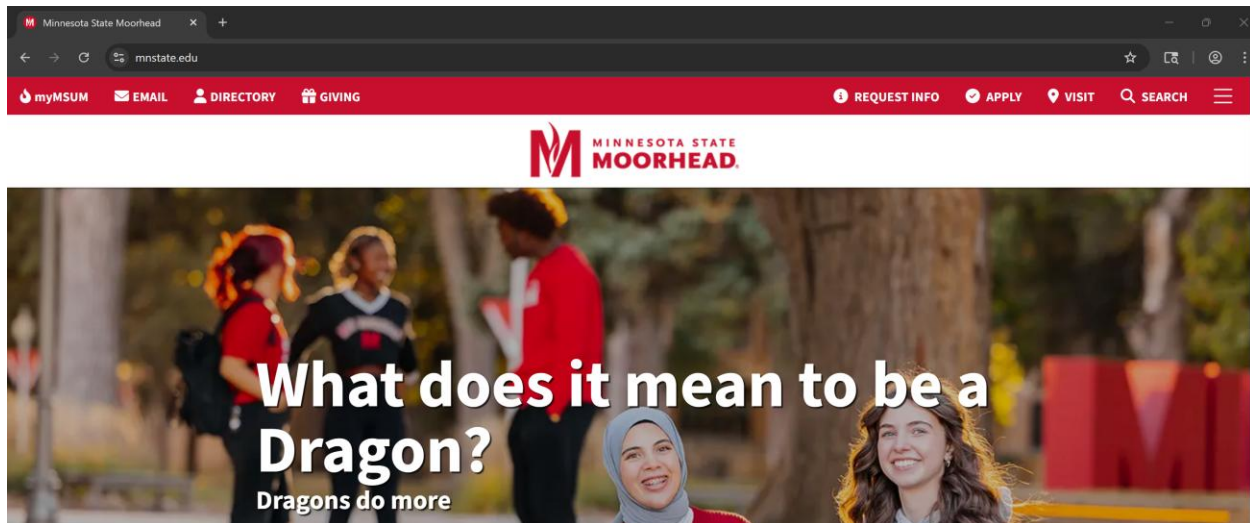


6. Changed the hosts files to its original status and save the changes.

```
hosts
File Edit View H1 ≡ B I ↺ ...
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97      rhino.acme.com      # source server
#       38.25.63.10     x.acme.com         # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1        localhost
#       ::1              localhost
# Added by Docker Desktop
10.0.0.4 host.docker.internal
10.0.0.4 gateway.docker.internal
# To allow the same kube context to work on the host and the container:
127.0.0.1 kubernetes.docker.internal
# End of section
```

7. After saving the changes to its original status, access to www.minstate.edu and browser reach to Minnesota State University Moorhead website.



Conclusion

This lab helps to cover the importance of DNS awareness and provides practical experience with a DNS poisoning attack scenario. The *nslookup* utility was used to perform basic DNS queries and domains resolve IP addresses. It shows the difference between centralized and distributed hosting models. The *dig* utility provided more details including DNS header information, recursion flags, TTL values, and message sizes, which are critical for diagnosing DNS misconfigurations and detecting abnormal behavior. DNS Poisoning demonstrated how DNS resolution can be overridden without requiring access to external DNS infrastructure.

The *nslookup* and *dig* utilities provide detailed information about DNS queries and resolution processes. Understanding DNS resolution paths and interpreting DNS query responses helps cybersecurity engineers to detect suspicious traffic patterns, identify DNS-based attacks, and troubleshoot resolution failures. The Domain Name System (DNS) server or the local host table can be an attack vector. Attackers can alter the local hosts file to create new entries such that users are redirected to a fraudulent site. As a cybersecurity engineer, this lab helps to analyze network behavior, recognize DNS-related threats, and implement defensive measures to mitigate risks associated with DNS exploitation.

References

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6. Amazon.com, Inc. (n.d.). *Amazon*. <https://www.amazon.com/>